

Scalability & Future Plan

Date	4 july 2026
Team ID	SWTID-2026-5594
Project Name	Credit Card Approval Prediction System
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority
1	The application currently runs on a local Flask server and is not designed for many simultaneous users.	Limited scalability	High
2	The dataset is highly imbalanced, which can bias predictions toward approvals.	Reduced prediction quality for minority cases	High
3	The system currently uses CSV files instead of a database.	Limited data management and reporting capabilities	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports local testing and small-scale usage	Deploy on a cloud platform (IBM Watson, Render, Azure, or AWS) with a production-ready WSGI server to handle more concurrent users.
2	Data Storage	CSV files are used for training data	Store application and prediction data in a relational database such as MySQL or PostgreSQL.
3	Performance	Model loads and predicts on a single server	Optimize the model, enable caching, and use load balancing if traffic increases.
4	Security	Basic form validation	Add HTTPS, authentication, authorization, secure configuration management, and input validation.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Deploy the application on IBM Watson Machine Learning or another cloud platform	1 Month	Enables online access and cloud-based predictions
Phase 2	Improve model performance using techniques such as SMOTE, hyperparameter tuning, and feature engineering	2 Months	Better prediction quality, especially for rejected applications
Phase 3	Integrate a database to store applicant information and prediction history	3 Months	Improved data management, reporting, and analytics
Phase 4	Add user authentication, an admin dashboard, analytics, and email/SMS notification support	4–6 Months	More secure, scalable, and production-ready application